

Heat Transfer Innovation Project

TOPIC: Transient Thermodynamic Analysis and Boundary Layer Optimization of a Thermoelectric Generator (TEG)



Objective

01

To compare the performance of a Thermoelectric Generator (TEG) under **natural** (no fan) and **forced** (with fan) convection. We aim to do this by analyzing the **temperature difference** created across the Peltier chips and measuring the resulting voltage.



Background

02

TEGs convert heat into electricity, but they need the **cold side to stay cold** to work efficiently. In still air, heat gets trapped around the cooling fins, which lowers the temperature difference and stops the module from making good power.



Problem Statement

03

How much does actively blowing air over the fins (forced convection) improve the **heat transfer rate** and the final **voltage output** compared to just leaving the setup in still air?

Experimental Setup & Apparatus

Technical specifications and physical configuration of the TEG system

Heat Source (Hot Boundary)

Transient cooling of boiling water

Reservoir	Steel container
Dimensions	17cm × 10cm × 5cm

Thermoelectric Module

Peltier TEG Chips

Footprint	120mm × 80mm
Type	Peltier TEG

Heat Sink (Cold Boundary)

Convective cooling via fins

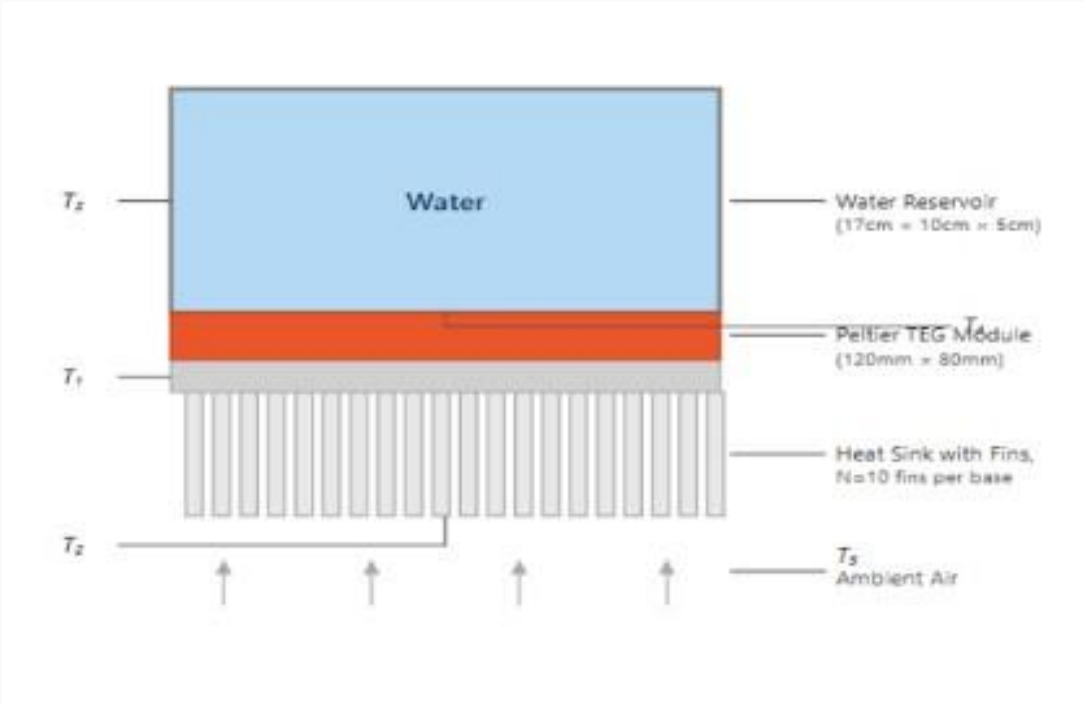
Base Dimensions	9cm × 4.7cm (3 bases)
Fin Geometry	L=1.6cm, W=0.1cm
Configuration	N=10 fins per base

Sensor Placement (Thermocouples)

T_1 - T_10 temperature monitoring

T_1 Fin Base	T_2 Fin Tip	T_3 Water	T_4 Cont. Base	T_5 Amb. Air
T_6 Top Lid	T_7 Side 1	T_8 Side 2	T_9 Side 3	T_10 Side 4

TEG System Cross-Section View



T_1 - T_10: Temperature monitoring points



Innovative Aspect

Active Cooling Comparison

Instead of just a standard static setup, we actively changed how heat escapes the system. By running the experiment both **with a fan (forced convection)** and **without a fan (natural convection)**, we could pinpoint exactly how much the trapped air was holding back the performance.

Real-Time Heat Tracking

Instead of using a constant electric heater, we used a container of **hot water**. By continuously measuring how fast the water cooled down over time, we were able to calculate the exact, real-time heat flow passing through the TEG chips.

Thermodynamic Circuit Modeling

We went beyond just measuring the electrical output. By applying basic **heat transfer calculations**, we showed that the main reason the system struggles to make more power is because **still air is terrible at absorbing heat** from the fins.

Overall, this setup acts as a practical, miniature model of how we can actually recover waste heat from hot liquids in real industrial plants.

Thermal Initialization

02

Heat transfers directionally from the hot reservoir base, through the thermoelectric (Peltier) module, and into the finned heat sink, establishing the temperature gradient (ΔT) required for voltage generation.

03

The transient cooling process is evaluated under two distinct convective states:

- 04

As the water reservoir naturally cools, the system continuously logs the temperature distribution across all sensors (T_1 to T_2) alongside the real-time voltage output generated across the Peltier chips.

Video



Heat Transfer Calculations

Finding the Heat Loss

$$Q_{total} = -m \cdot C_p \cdot \frac{dT_3}{dt}$$

$$Ra = \frac{g \cdot \beta \cdot \Delta T \cdot L_c^3}{\nu \cdot \alpha_{air}}$$

$$h = \frac{Nu \cdot k_{air}}{L_c}$$

We first calculated the total heat leaving the cooling water. To be accurate, we didn't just guess the heat lost to the room—we calculated the actual convective heat transfer coefficient (h) for the sides and top of the box using standard Rayleigh (Ra) and Nusselt (Nu) number formulas.

Fourier's Law of Conduction

$$Q_{TEG} = \frac{k_{TEG} \cdot A \cdot (T_4 - T_1)}{H_{TEG}}$$

After subtracting the room losses, we used Fourier's Law to calculate the exact amount of heat flowing straight down through the Peltier chips. This depends on the measured boundary temperatures, along with the chips' thermal conductivity (k) and thickness (H).

Voltage vs. Temperature

$$V = \alpha \cdot \Delta T$$

We plotted the generated voltage against the temperature difference (ΔT) across the chips to calculate the experimental Seebeck coefficient (α) for both cases:

Without Fan: 0.0889

With Fan: 0.1113

Properties Used

- Kinematic Viscosity of Air (ν): $1.51 \times 10^{-5} \text{ m}^2/\text{s}$
- Thermal Diffusivity of Air (α_{air}): $1.90 \times 10^{-5} \text{ m}^2/\text{s}$
- Thermal Conductivity of Air (k_{air}): $0.026 \text{ W/m}\cdot\text{K}$
- Thermal Conductivity of TEG (k_{TEG}): $0.1815 \text{ W/m}\cdot\text{K}$

Sample Calculations

Case 1: Without air:

Consider the first reading:

T_1 (Fin Base Temperature) = 47.0 °C

T_2 (Fin Tip Temperature) = 42.0 °C

T_3 (Water Temperature) = 82.0 °C

T_4 (Container Base Temperature) = 69.5 °C

T_5 (Ambient Air Temperature) = 32.0 °C

T_6 (Top Lid Temperature) = 62.9 °C

T_7 (Temperature of vertical back surface of the box) = 68.0 °C

T_8 (Temperature of vertical front surface of the box) = 67.1 °C

T_9 (Temperature of vertical left surface of the box) = 67.5 °C

T_{10} (Temperature of vertical right surface of the box) = 66.5 °C

$V = 2.0 \text{ V}$

Sample Calculations

Given Constants:

L_{box} (Length of Box) = 0.17 m

W_{box} (Width of Box) = 0.10 m

H_{box} (Height of Box) = 0.05 m

L_{peltier} (Side length of Peltier chip) = 0.04 m

C_{water} (Specific Heat of Water) = 4186.5 J/mol K

ρ_{water} (Density of Water) = 986 kg/m³

ν_{air} (Kinematic Viscosity of Air) = $1.51 \cdot 10^{-5}$ m²/s

α_{air} (Thermal Diffusivity of Air) = $1.9 \cdot 10^{-5}$ m²/s

k_{air} (Thermal Conductivity of Air) = 0.026 W/m K

g (Acceleration due to Gravity) = 9.81 m/s²

V_{water} (Volume of water in the box) = $2.20 \cdot 10^{-4}$ m³

k_{peltier} = 0.1815 W/m K

H_{peltier} = 10^{-3} m

Sample Calculations

Seeback Coefficient Calculation:

ΔT = Temperature difference across Peltier chips = $T_4 - T_1 = 69.5 - 47.0 = 22.5$ °C

α (Seeback Coefficient) = $V/\Delta T = 2.0/22.5 = 0.0888$ V/°C

Similarly, Seeback Coefficient for other cases are also calculated and the averaged to get the mean Seeback Coefficient of the Peltier chips.

$$\alpha_{\text{mean}} = \langle \alpha \rangle = 0.0889$$

Sample Calculations

Thermal Calculations:

$$m_{\text{water}} = \rho_{\text{water}} * V_{\text{water}} = 0.21692 \text{ kg}$$

dT_3/dt is found by fitting a straight line to the graph of T_3 vs t

$$dT_3/dt = -0.0678 \text{ } ^\circ\text{C/s}$$

$$A_{\text{top}} = A_6 = L_{\text{box}} * W_{\text{box}} = 0.0170 \text{ m}^2$$

$$A_{\text{back}} = A_{\text{front}} = A_7 = A_8 = L_{\text{box}} * H_{\text{box}} = 0.0085 \text{ m}^2$$

$$A_{\text{left}} = A_{\text{right}} = A_9 = A_{10} = W_{\text{box}} * H_{\text{box}} = 0.005 \text{ m}^2$$

$$T_{\text{film } 6} = (T_6 + T_5)/2 = 47.45$$

$$T_{\text{film } i} = (T_i + T_5)/2 = [50, 49.55, 49.75, 49.25] \quad i=7, 8, 9, 10$$

$$\beta_6 = 1/T_{\text{film } 6} = 0.0211$$

$$\beta_i = 1/T_{\text{film } i} = [0.02, 0.0202, 0.0201, 0.0203] \quad i=7, 8, 9, 10$$

$$Ra_6 = (g * \beta_6 * (T_6 - T_5) * (A_6 / (2 * (L_{\text{box}} + W_{\text{box}}))))^3 / (v_{\text{air}} * \alpha_{\text{air}}) = 1.0283 * 10^5$$

$$Ra_i = (g * \beta_i * (T_i - T_5) * H_{\text{box}}^3) / (v_{\text{air}} * \alpha_{\text{air}}) = [4.7615, 4.649, 4.699, 4.5738] * 10^5 \quad i=7, 8, 9, 10$$

$$Pr = v_{\text{air}} / \alpha_{\text{air}} = 0.7947$$

$$Nu_i = (0.825 + 0.387 * Ra_i^{(1/6)} / (1 + (0.492 / Pr)^{(9./16)}))^{(8/27)} = [13.8061, 13.7207, 13.7589, 13.6629]$$

Sample Calculations

if $Ra_6 > 10^4$ and $Ra_6 < 10^7$

$$Nu_6 = 0.54 * Ra_6^{0.25}$$

If $Ra_6 > 10^7$

$$Nu_6 = 0.15 * Ra_6^{(1/3)}$$

In this reading, $Ra_6 > 10^4$ and $Ra_6 < 10^7$

$$Nu_6 = 0.54 * Ra_6^{0.25} = 9.6698$$

$$h_i = Nu_i * k_{air} / H_{box} = [7.1792, 7.1348, 7.1546, 7.1047] \text{ W/m}^2 \text{ K} \quad i=7,8,9,10$$

$$h_6 = Nu_6 * k_{air} / (A_6 / (2 * (L_{box} + W_{box}))) = 7.9861 \text{ W/m}^2 \text{ K}$$

$$Q_i = h_i * A_i * (T_i - T_5) = [2.1968, 2.1287, 1.2699, 1.2256] \text{ W} \quad i=7,8,9,10$$

$$Q_6 = h_6 * A_6 * (T_6 - T_5) = 4.1951 \text{ W}$$

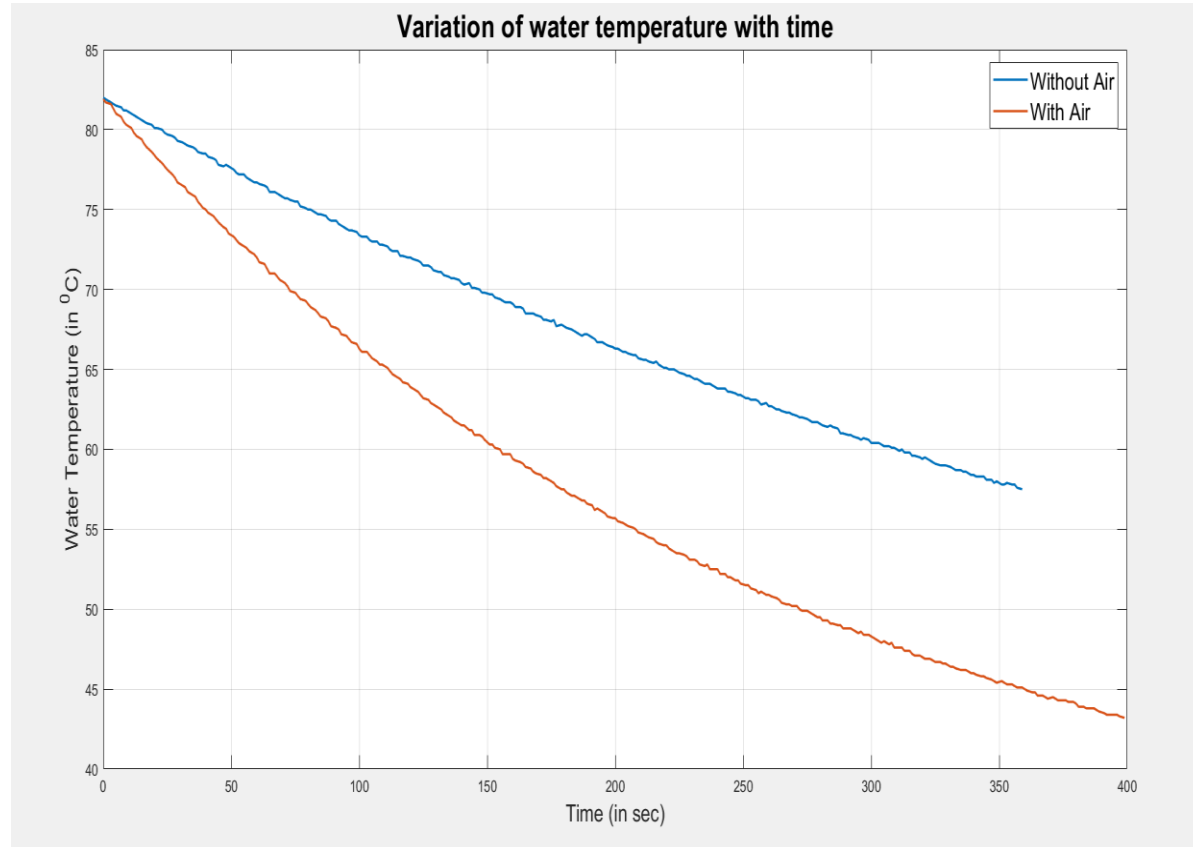
$$Q_{total} = m_{water} * C_{water} * dT_3 / dt = 61.2050 \text{ W}$$

$$Q_{bottom} = Q_{total} - \sum Q_i = 50.1889 \text{ W}$$

Sample Calculations

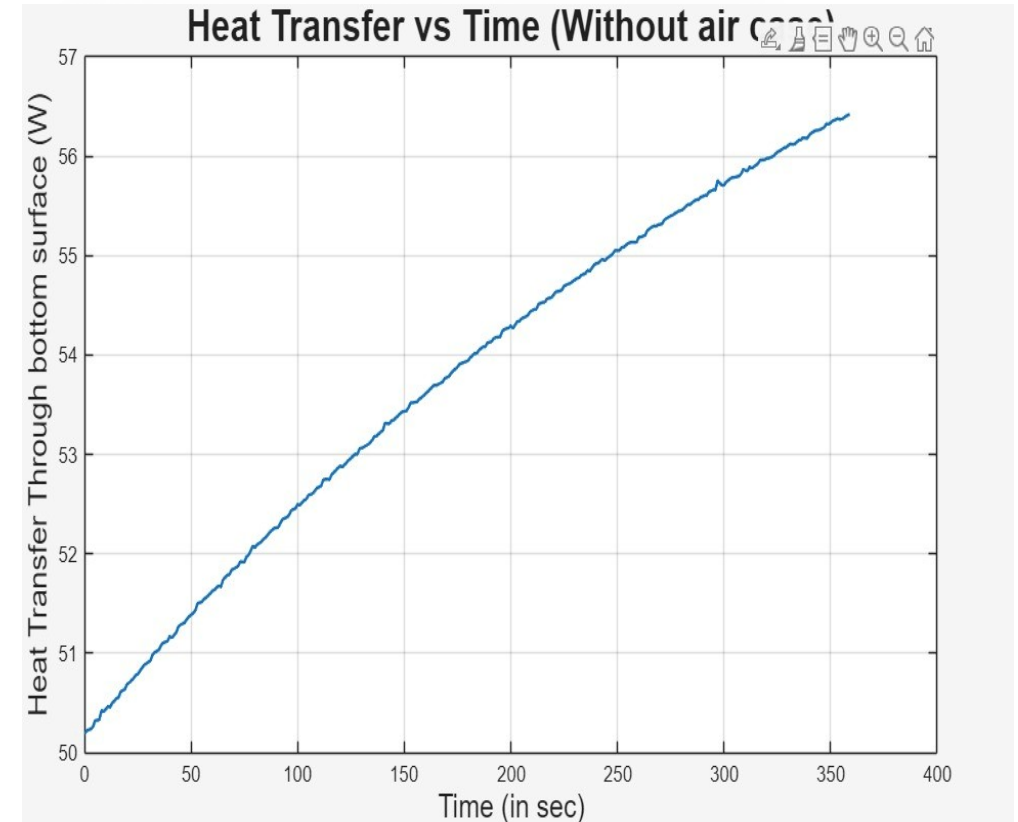
$$Q_{\text{peltier}} = -k_{\text{peltier}} * A_{\text{peltier}} * (T_4 - T_1) / H_{\text{peltier}} = 39.2040 \text{ W}$$

Results I: Transient Cooling & Heat Extraction



The Cooling Curve (dT/dt)

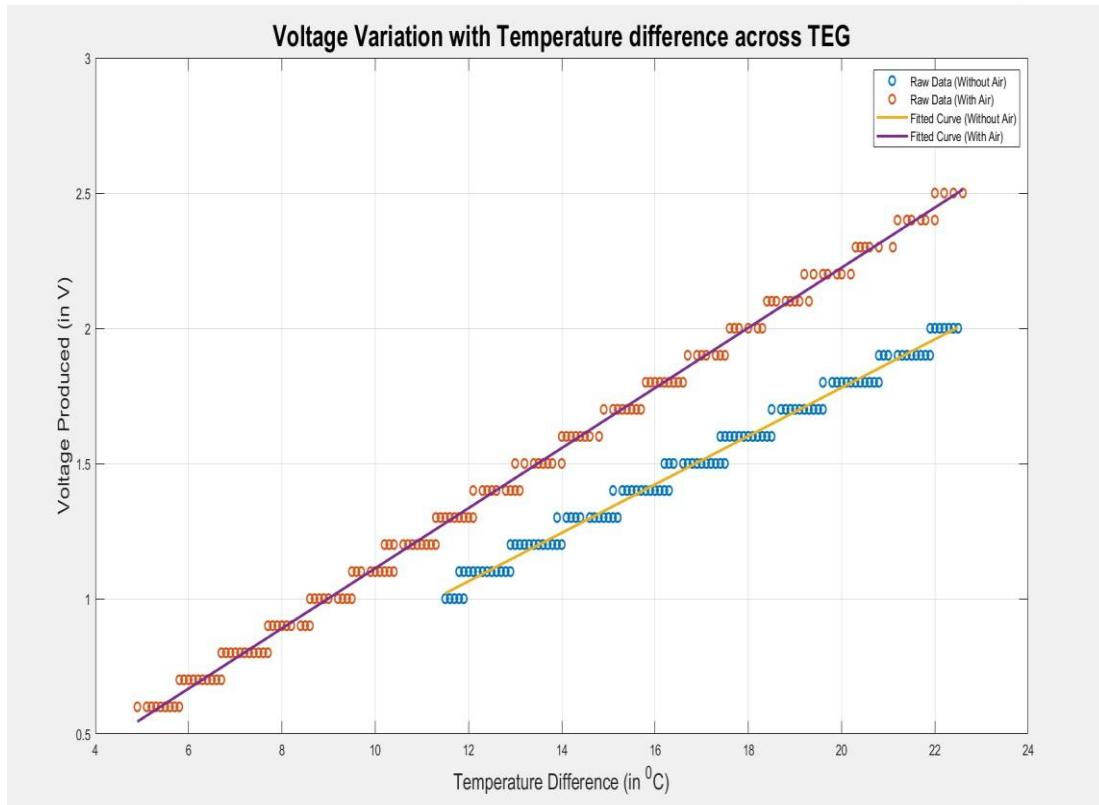
The experimental data demonstrates a significantly steeper negative slope for the water reservoir temperature under forced convection ("With Air"). This proves that actively disrupting the boundary layer mechanically forces the thermal mass to shed heat at an accelerated rate.



Heat Flux

Heat transfer through bottom of container without flow of air

Results II: Thermoelectric Generation & Seebeck Shift



Validation of the Seebeck Effect

Both experimental curves confirm the fundamental principle of thermoelectricity: generated voltage is directly proportional to the macroscopic temperature gradient (ΔT).

The "Forced Convection" Shift

For any given ΔT , the "With Air" state produces a higher voltage output. The slope is steeper, and the data is shifted upward.

Physical Justification

The doubled heat flux (proven in the previous slide) prevents thermal stagnation inside the ceramic module. This maintains a much sharper, high-fidelity microscopic temperature gradient directly across the internal P-N semiconductor junctions, maximizing electrical conversion.

Analysis: The Cooling Problem

The biggest problem with generating power from a TEG isn't the chip itself — it's how hard it is to get the heat out of the cooling fins into the room.



Without a Fan (Natural Convection)

In still air, the heat transfer coefficient (h) is very low. The heat gets trapped, causing the cold-side fins to heat up rapidly. This shrinks the temperature difference (ΔT) across the chips, resulting in lower voltage output.



With a Fan (Forced Convection)

Turning on the fan actively blows the trapped hot air away from the fins. As our sensor data showed, this cools the fin base (T_1) much faster. This maintains a wider temperature difference (ΔT) across the module, generating higher voltage.



Final Conclusion

To get the most electricity out of a TEG, the cold side must be kept as cold as possible. Actively improving the cooling on the exhaust side (increasing h) is the most effective way to maximize the heat flow passing through it.

Conclusion

Below are the core conclusions drawn from this experiment:

- In this experiment, we tested practical ways to **reduce thermal resistance** and improve how a TEG actually performs.
- By comparing the setup with and without a fan, we showed that **forced convection** is the best way to increase the heat flow and get more voltage.
- We ran physical experiments using hot water and backed up our data with **MATLAB energy calculations** to find exactly where the heat was getting trapped.
- Our setup worked well as a small-scale model for **real-world waste heat recovery**, using hot water as the heat source and a finned heat sink for cooling.
- Overall, this project helped us connect classroom **heat transfer concepts** to a real, physical system and prove it with our own data.

Experimental Constraints & Sources of Error



Non-Ideal Heat Source

The steel container we used had some sealed holes at the bottom, which caused minor leaks and an uneven surface. This means our heat flow wasn't perfectly 1-D, but the overall effect on the data was small.



Sensor Placement

Because the setup was tightly clamped together, we couldn't place the thermocouples exactly at the perfect theoretical boundaries. They were slightly shifted, which causes tiny offsets in the raw temperature readings.



Uneven Airflow

The fan we used blew air mostly at the center of the heat sink. The air didn't spread perfectly across all the outer fins, meaning the cooling wasn't 100% uniform across the entire cold side.

Conclusion

These are minor constraints. Our overall results and the clear trend between the natural and forced convection cases are still completely valid. Fixing these issues would only slightly optimize the final numbers, not change the core findings.

Future Optimization & Industrial Applications

Future Experimental Work

Minimizing Thermal Contact Resistance (R_{contact}):

Implementing high-conductivity thermal paste or graphite pads to eliminate insulating air gaps between the TEG, heat sink, and reservoir, thereby maximizing the total heat flux (Q).

Enforcing 1-D Heat Conduction: Utilizing a custom-machined, vacuum-insulated thermal reservoir to completely eliminate fluid leakage and parasitic ambient heat losses.

High-Resolution Data Acquisition: Embedding micro-thermocouples directly into the metallic interfaces to capture the exact mathematical boundary temperatures without spatial offset errors.

Industrial Scalability (Waste Heat Recovery)

Exploiting Parasitic Thermal Losses: Industrial machinery, chemical reactors, and exhaust pipelines generate massive amounts of continuous, unutilized heat. TEGs can tap into this "free" thermal gradient.

Auxiliary Power Generation: By applying the optimized forced-convection boundary layer principles tested in this project, industrial plants can convert waste heat into usable electrical power.

Self-Sustaining Sensor Networks: The generated electricity ($\sim 1\text{ V} - 5\text{ V}$ scaled) is perfectly suited to power decentralized industrial IoT (Internet of Things) sensors and wireless monitoring systems directly at the equipment level, completely independent of the main power grid.

Thank You

We welcome any questions, feedback, or discussion regarding the methodology and results.